LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	27
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Maintenance of O2 system, servicing and safety precautions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Maintenance of O2 system, servicing and safety precautions. Safety measures while handling /upkeep of gaseous oxygen system. Quality checks on pressurized gas cylinders.
5. Specific Objectives	23.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of O2 cylinder quality checks, safety valve inspection, and leak detection procedures with instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary safety risks associated with gaseous oxygen?
(b) List three common types of oxygen systems used in aviation.
(c) Why is oil contamination strictly prohibited in oxygen systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need and types of oxygen systems • Working principles of O2 components • Safety measures for gaseous oxygen handling	L & Ds	PPT, IB	15 min
(b)	• Quality checks on pressurized gas cylinders • Servicing procedures and documentation • Inspection of safety valves and regulators	L & Ds	PPT, PDF	20 min
(c)	• Leak detection techniques • Storage and transport safety protocols • Emergency response to O2 system failures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct procedure for checking the hydrostatic test date on an oxygen cylinder.
(b) Describe the immediate actions required if an oxygen leak is detected in the cabin.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between gaseous and liquid oxygen systems, focusing on maintenance requirements, to be discussed in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Typical cabin pressurization system, Cabin pressure regulator safety v/v, outflow v/v etc and its operation, Methods of sealing of cabin	Refer to the master syllabus reference list	LP 28

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min